

IT353 BLOCKCHAIN TECHNOLOGY



Dr. Shamama Anwar

Assistant Professor

Department of Computer Science and Engineering,

BIT, Mesra

shamama@bitmesra.ac.in

9431103880

Course Outcomes

After the completion of this course, students will be able to:

1. Learn and explain the difference between centralized, decentralized network and blockchain.
2. Explain fundamental concepts of blockchain using hashes and consensus.
3. Understand the concept of mining in blockchains.
4. Understand the working of Bitcoin and its security.
5. Know about the different platforms for implementing blockchain and its varied application.

Syllabus

MODULE – I

- **Introduction to Blockchain Technology:** Introduction to Blockchain, Trusted Third party for transactions, Difference between centralized, decentralized and distributed peer to peer networks, Types of Blockchain (Permission Blockchain vs. Permissionless Blockchain), History of Bitcoins.

MODULE – II

- **Fundamental concepts of Blockchain:** Concepts of Block, Transactions, Hashes, Consensus. Hashes: Hash cryptography, Encryption vs. hashing, Transactions: Recording transactions, Digital Signature, Verifying and confirming transactions, Blocks and blockchain: Hash pointers, Blocks, Consensus building. Distributed consensus, Byzantine generals problem, Consensus mechanism: POW, POS, POB, POA, etc. Blockchain Architecture, Markle Root Tree.

MODULE – III

- **Mining and simulating blockchain:** Mining and simulating blockchain: Game theory behind competitive mining. Incentives: mining and transaction fees, Energy expended in mining.

Syllabus

MODULE – IV

- **Bitcoin and Security:** Bitcoin creation, exchanges. Wallets, security. Protecting blockchain from attackers. Forks soft and hard, Blockchain security, Key Management in Bitcoin, Case studies.

MODULE – V

- **Platforms and Applications:** Introduction to Blockchain platform: Ethereum, Hyperledger, IOTA, EOS, Multichain, SOLIDITY, Designing a new blockchain, Distributed Application (DAPP).
- **Applications:** E-Governance, Elections, File sharing, Micropayments
- Challenges and Research Issues in blockchain

Books

Text Book :

- Bitcoin and Cryptocurrency technologies: A comprehensive introduction. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder. Princeton University Press, First edition, 2016
- Blockchain Applications: A Hands-On Approach. Arshdeep Bahga, Vijay Madisetti. VPT Publisher. First edition, 2018.
- Blockchain: Step by Step Guide to Understand by Paul Laurence, Createspace Independent Pub.

Reference Book:

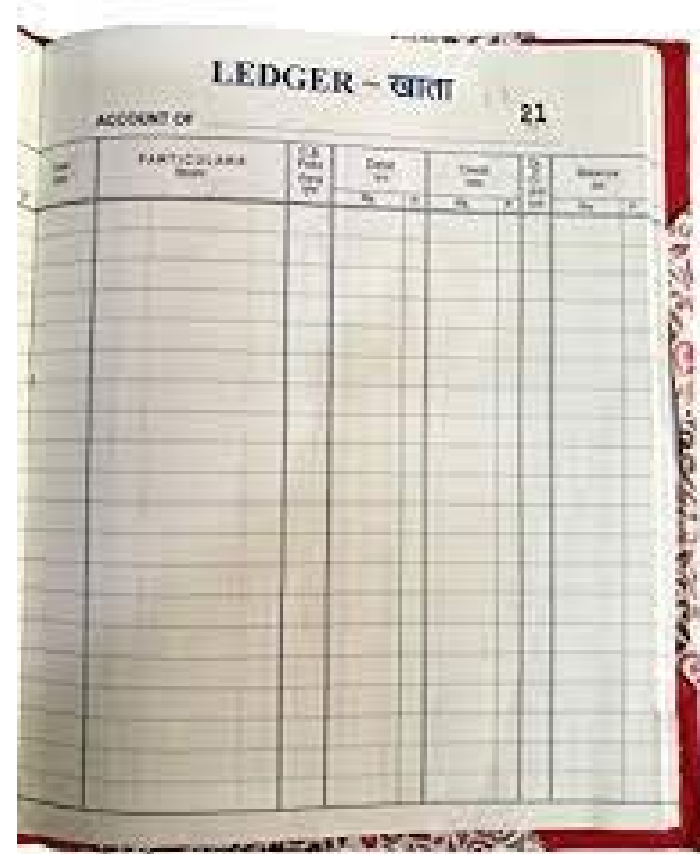
- Introducing Ethereum and Solidity Foundations of Cryptocurrency and Blockchain Programming for Beginners by Chris Dannen, Apress
- Blockchain: The Comprehensive Beginner's Guide by Frank Walrтин

Module - I

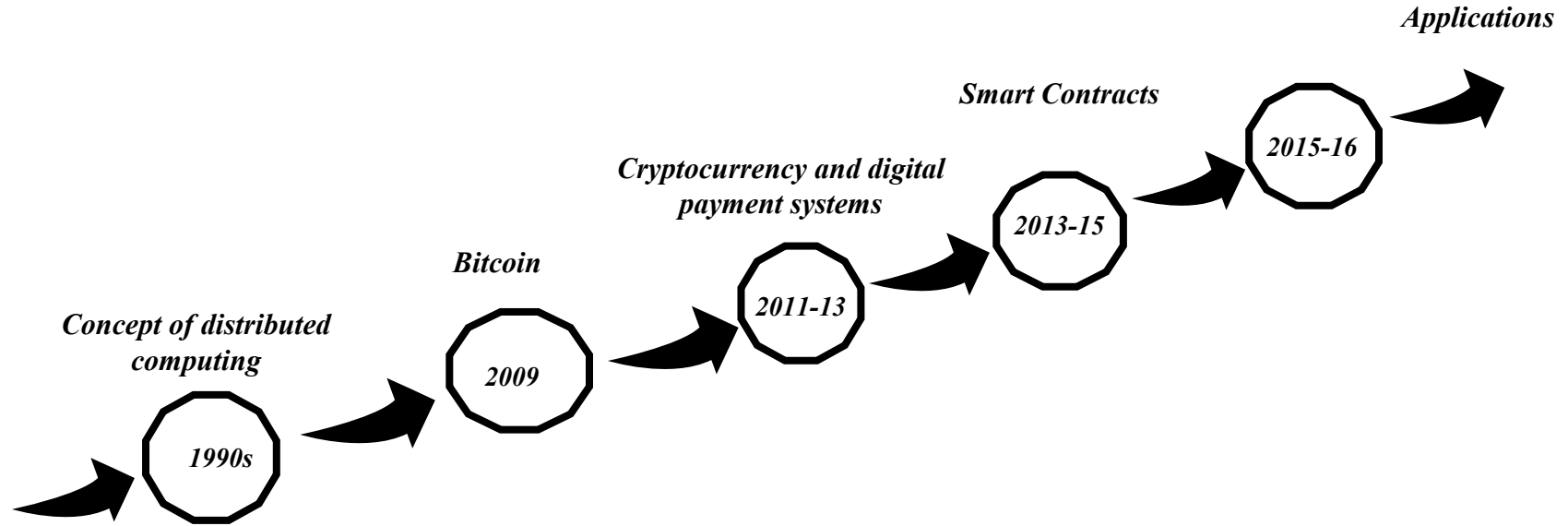
- Introduction to Blockchain Technology
- Trusted Third party for transactions
- Difference between centralized, decentralized and distributed peer to peer networks
- Types of Blockchain (Permission Blockchain vs. Permissionless Blockchain)
- History of Bitcoins.

Introduction to Blockchain

- Blockchain are often confused with cryptocurrencies specifically Bitcoins.
- **Blockchain** is a decentralized ledger, tracking digital assets on a Peer-to-Peer network.
- A **ledger** is a record book, which holds details of every transaction that takes place on the network.
- Each system has a copy of this ledger, hence this ledger is **distributed**.



Evolution of Blockchain



Advantage of Blockchain

- Immutability*
- Data sharing & Access*
- Decentralized Storage*
- Security*
- Fraud Detection*
- Scalability*
- Authentication*
- Privacy*
- Distributed & Transparent*

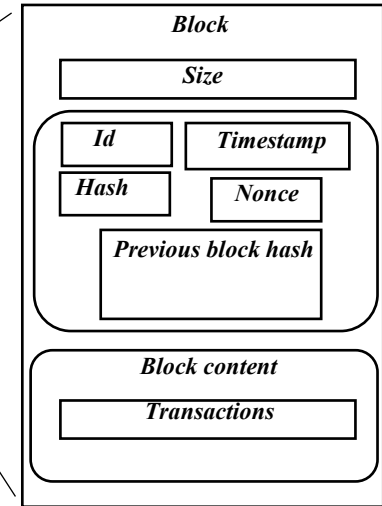
Evolution of Blockchain



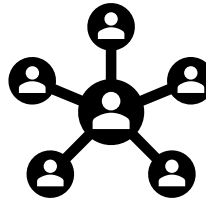
1 *User requests transactions*



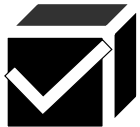
2 *A block representing the transaction is created*



3 *The block is broadcasted to all the nodes*



4 *All the nodes validate the block*



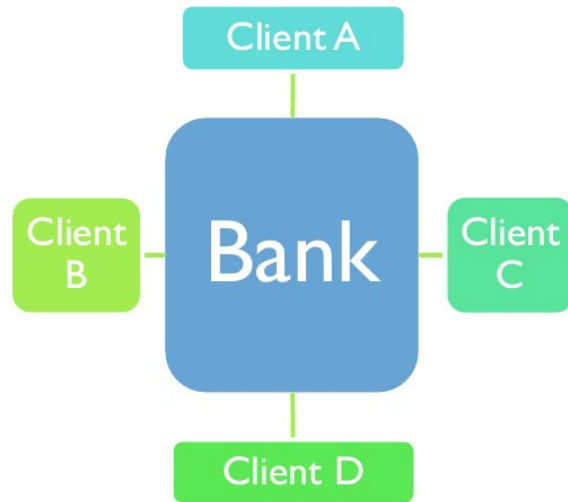
5 *The block is added to the chain*



6 *The transaction gets verified and executed*

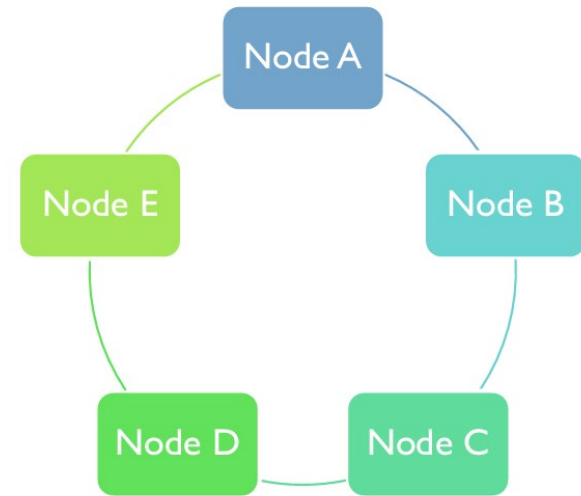
Introduction to Blockchain

- Centralized vs. Distributed ledger



Centralized ledger

- There are multiple ledgers but a central one is maintained for actual usage.
- Different clients must reconcile their ledger with the central ledger



Distributed ledger

- There is one ledger. All Nodes have some level of access to that ledger.
- All Nodes agree to a protocol that determines the "true state" of the ledger at any point in time (consensus).

Introduction to Blockchain

- **Distributed ledgers**

- Can be used without a central authority or third party intervention.
- Can be used to create value and issue assets.
- Can be used to transfer value or ownership of assets.

- **Trusted Third Party (TTP)**

- A trusted third party (TTP) is an entity which facilitates interactions between two parties who both trust the third party.
- TTP reviews all critical transaction communications between the parties for approval.
- By greatly reducing paperwork and errors, blockchain significantly reduces overhead and transaction costs, and reduces or eliminates the need for third parties or middlemen to verify transactions.

Centralized vs. Decentralized vs. Distributed N/W

• Centralized System

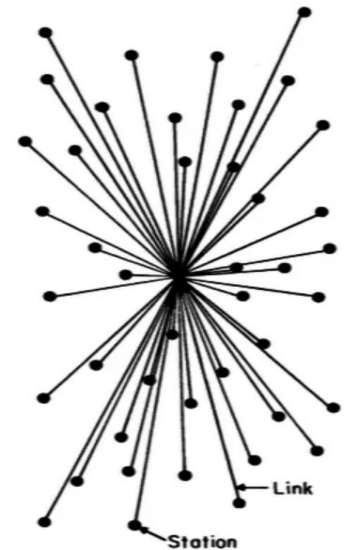
- All users are connected to a central network owner or “server”.
- The central owner stores data, which other users can access, and also user information

✓ Advantages

- Simple deployment
- Can be developed quickly
- Affordable to maintain
- Practical when data needs to be controlled centrally

× Disadvantages

- Prone to failures
- Higher security and privacy risks for users
- Longer access times to data for users who are far from the server



Centralized vs. Decentralized vs. Distributed N/W

• Decentralized System

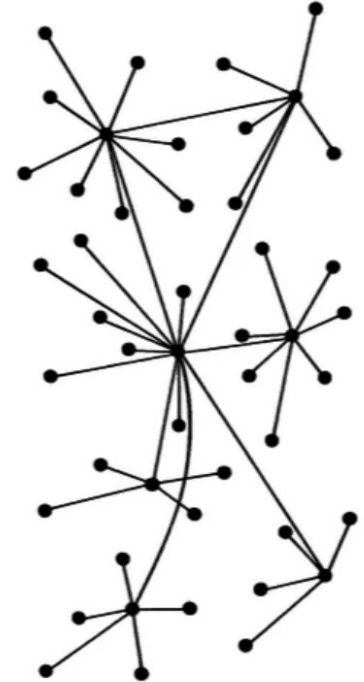
- Doesn't have one central owner. Instead, they use multiple central owners, each of which usually stores a copy of the resources users can access.
- Resources remain active if at least one of the central servers continue to operate. Usually, this means that system owners can repair faulty servers and address any other problems while the system itself continues to run as usual.

✓ Advantages

- Less likely to fail than a centralized system
- Better performance
- Allows for a more diverse and more flexible system

× Disadvantages

- Security and privacy risks to users
- Higher maintenance costs
- Inconsistent performance when not properly optimized



Centralized vs. Decentralized vs. Distributed N/W

• Distributed System

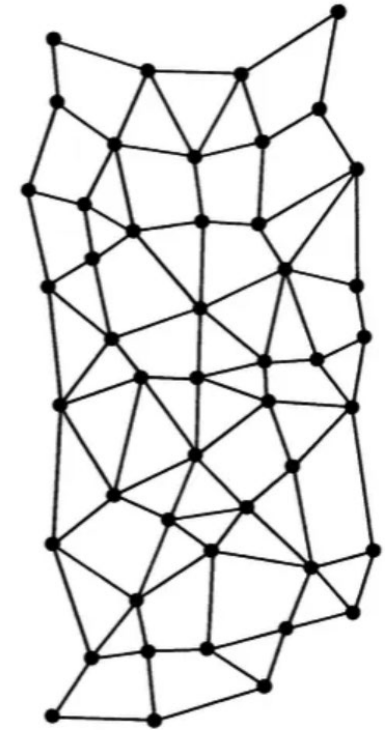
- Users have equal access to data, though user privileges can be enabled when needed.
- The distributed system enables users to share ownership of the data.
- Hardware and software resources are also allocated between users, which in some cases may improve the performance of the system.
- A distributed system is safe from the independent failure of components, which can improve its uptime considerably.

✓ Advantages

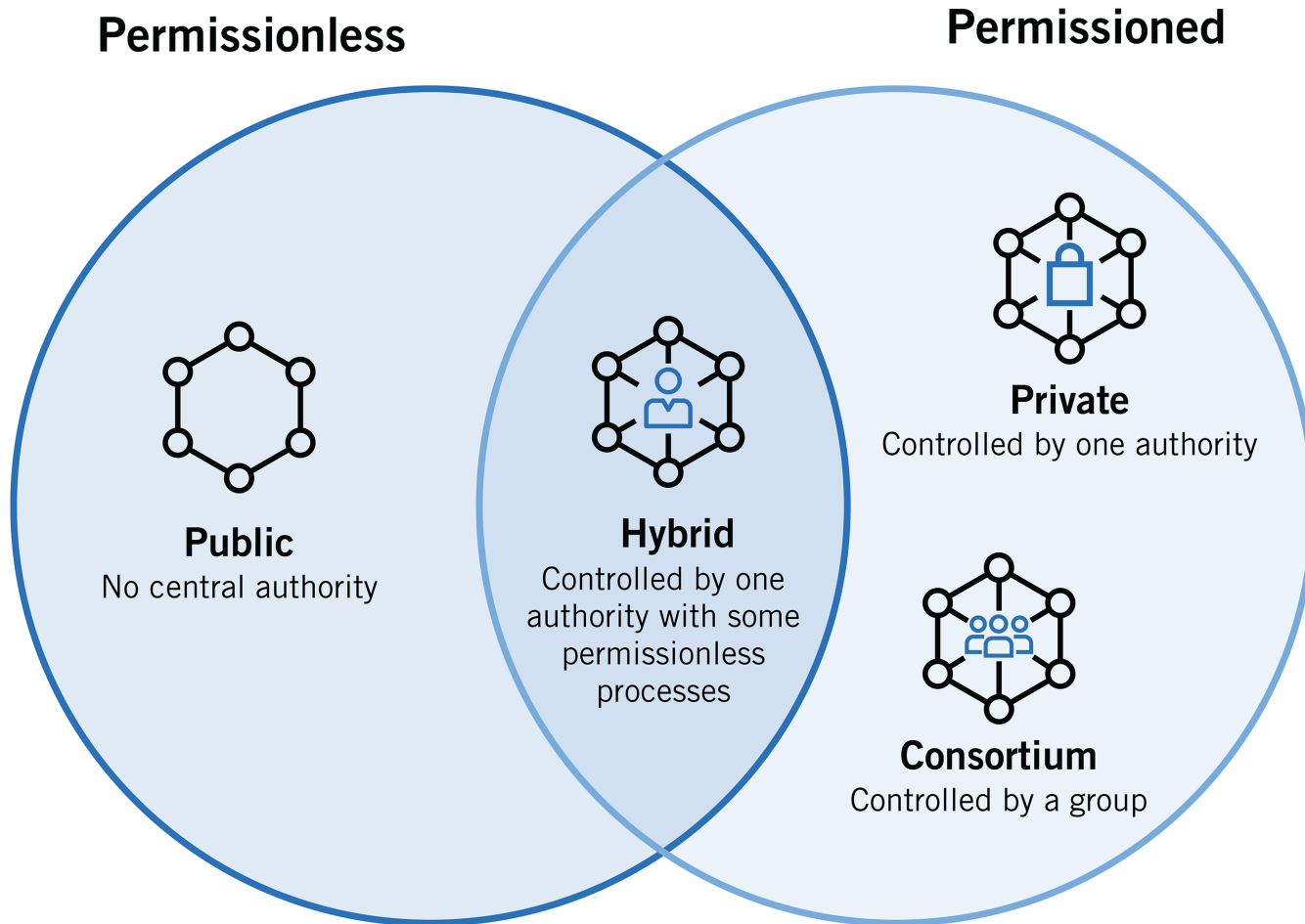
- Fault-tolerant
- Transparent and secure
- Promotes resource sharing
- Extremely scalable

✗ Disadvantages

- More difficult to deploy
- Higher maintenance costs



Types of Blockchain



Types of Blockchain

• Public

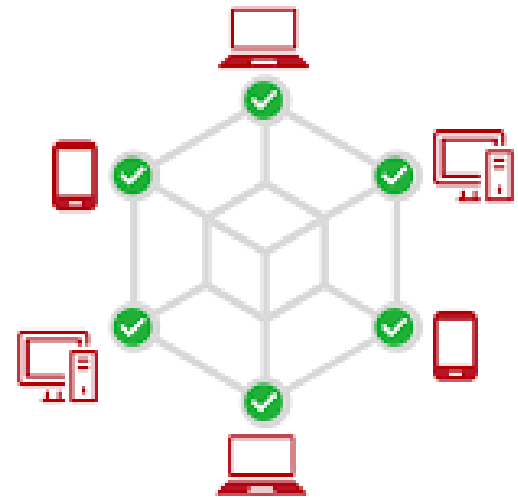
- Non-restrictive, permission-less distributed ledger system.
- Anyone who has access to the internet can sign in on a blockchain platform to become an authorized node and be a part of the blockchain network.
- A node or user which is a part of the public blockchain is authorized to access current and past records, verify transactions and do mining.
- The most basic use of public blockchains is for mining and exchanging cryptocurrencies.
- Example: Bitcoins, Ethereum

✓ Advantages

- Open and trusted

✗ Disadvantages

- Lower speed (TPS)
- Scalability Issue
- High Energy Consumption



Types of Blockchain

• Private

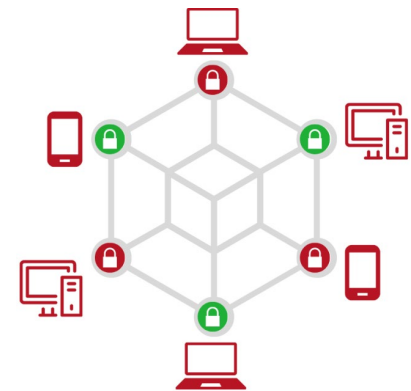
- A private blockchain is a restrictive or permission blockchain operative only in a closed network.
- Private blockchains are usually used within an organization or enterprises where only selected members are participants of a blockchain network.
- The level of security, authorizations, permissions, accessibility is in the hands of the controlling organization.
- Private blockchain networks are deployed for voting, supply chain management, digital identity, asset ownership, etc.
- Examples: Multichain and Hyperledger projects (Fabric, Sawtooth), Corda, etc.

✓ Advantages

- Higher speed (Transactions per second (TPS))
- Scalability

✗ Disadvantages

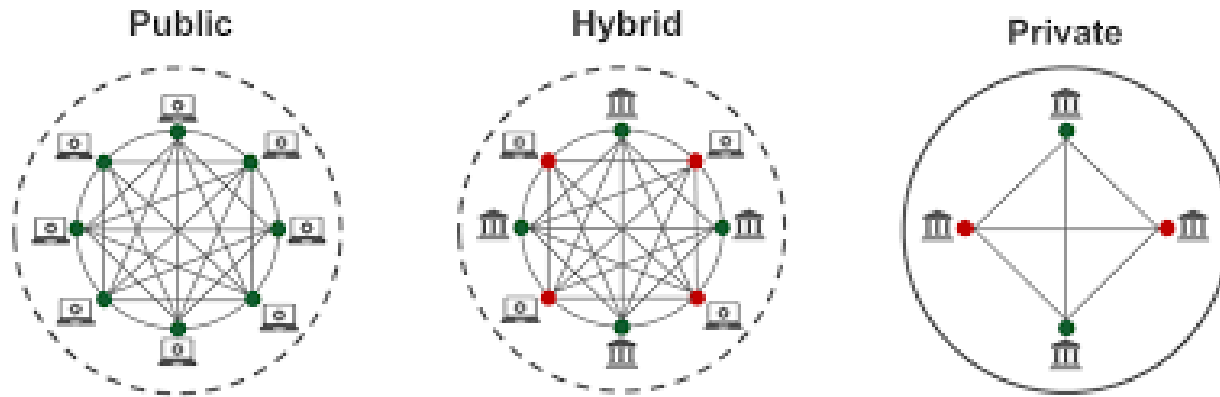
- Centralized



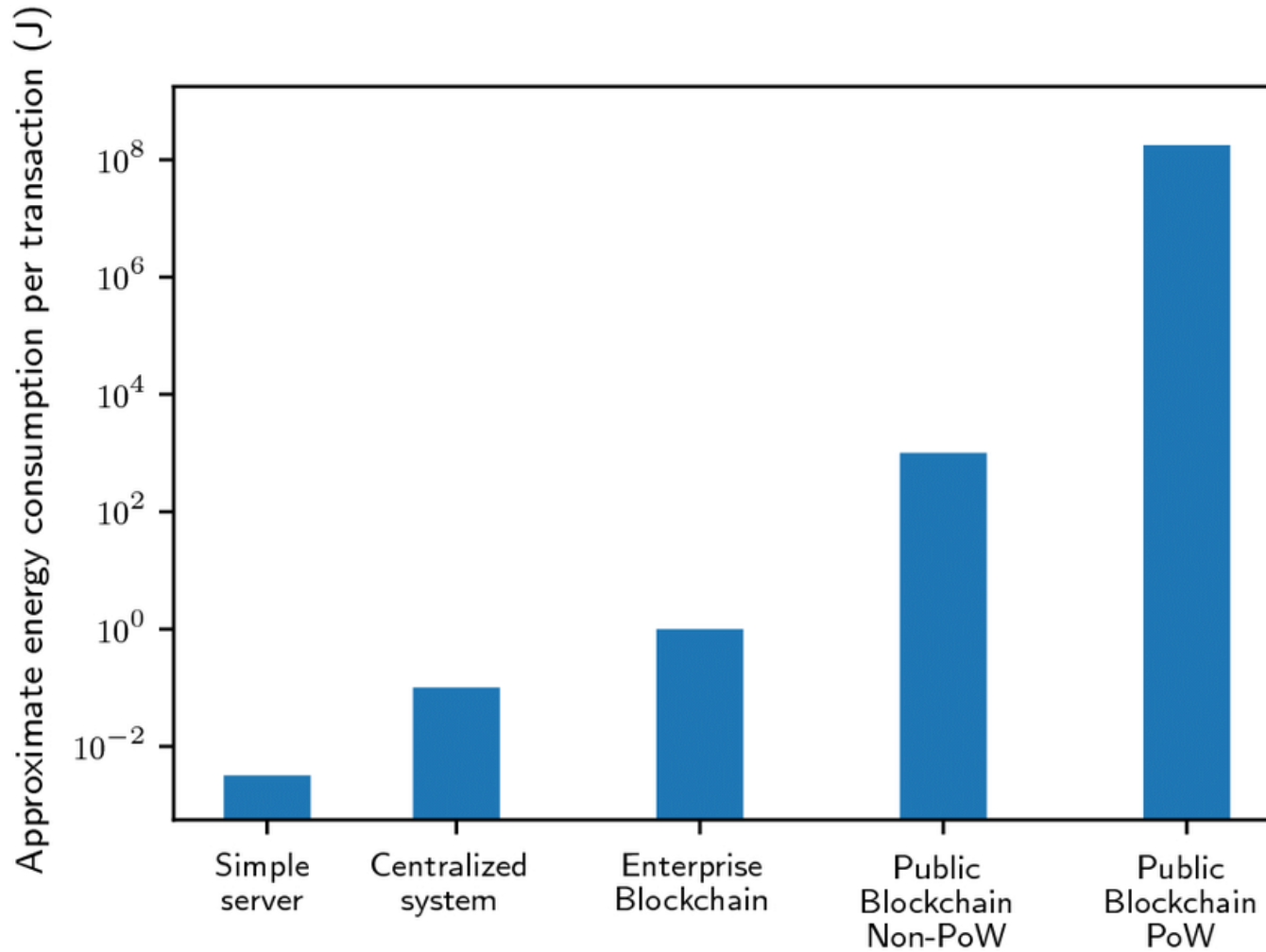
Types of Blockchain

- **Hybrid**

- A hybrid blockchain is a combination of the private and public blockchain. It uses the features of both types of blockchains that is one can have a private permission-based system as well as a public permission-less system.
- With such a hybrid network, users can control who gets access to which data stored in the blockchain. Only a selected section of data or records from the blockchain can be allowed to go public keeping the rest as confidential in the private network.
- Example: Dragonchain.



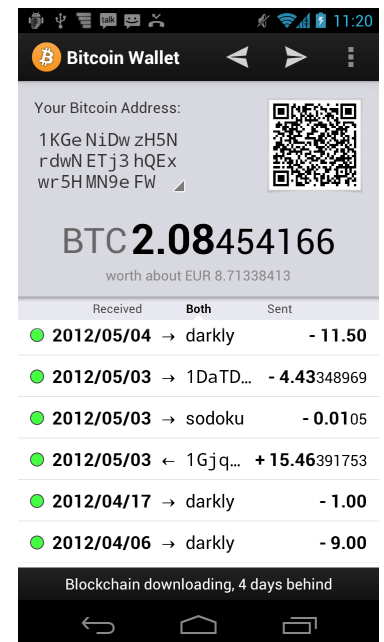
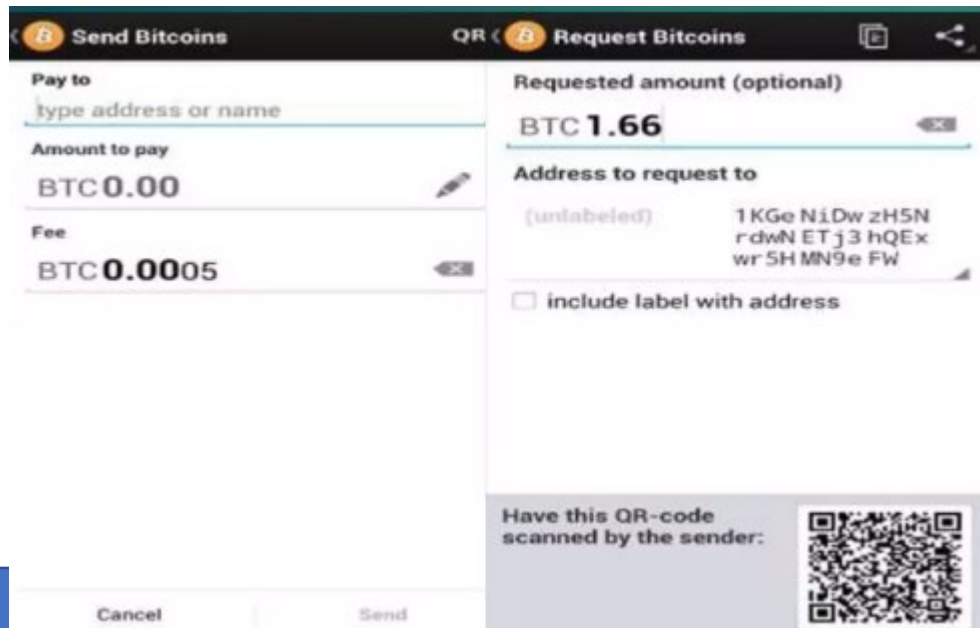
Energy Consumption in Blockchain



Sedlmeir, J., Buhl, H.U., Fridgen, G. *et al.* The Energy Consumption of Blockchain Technology: Beyond Myth. *Bus Inf Syst Eng* **62**, 599–608 (2020).
<https://doi.org/10.1007/s12599-020-00656-x>

Bitcoins

- Bitcoin is considered to be the first cryptocurrency and blockchain technology.
- Satoshi Nakamoto introduced the concept in 2008.
- It is a virtual currency (no physical form)
- Not regulated by any central banking system (decentralized) and its value varies based on demand.
- It's a public blockchain, hence everything on Bitcoin is visible to the users (transparent).
- The users work through an account (wallet).



Bitcoins

- At the backend it works on a distributed ledger that records all transactions.

The screenshot shows the Blockchain.info website. The top navigation bar includes links for Home, Charts, Stats, Markets, API, and Wallet. A search bar is located on the right. The main content area is titled 'Home Welcome to Blockchain' and features a table of recent transactions. The table has columns for Height, Age, Transactions, Total Sent, Relayed By, and Size (kB). Below the table, there is a 'Latest Transactions' section showing a transaction with a green status and a 'Search' section with a text input and a 'Search' button.

Height	Age	Transactions	Total Sent	Relayed By	Size (kB)
440828	8 minutes	2241	4,973.59 BTC	HaoBTC	998.14
440827	12 minutes	1918	14,536.52 BTC	Unknown	749.24
440826	30 minutes	247	514.39 BTC	BW.COM	109.99
440825	31 minutes	590	11,337.59 BTC	Bitcoin.com	998.83
440824	34 minutes	1425	3,926.33 BTC	BitClub Network	998.21
440823	38 minutes	2417	10,631.93 BTC	BW.COM	959.3

Latest Transactions

05b7cc6c90a475cf93270e18... < 1 minute 0.5414958 BTC

Search
You may enter a block height, address, block hash, transaction hash, hash160, or ipv4 address...

Address / ip / SHA hash

<https://www.slideshare.net/kashif2016/what-is-bitcoin-its-history-and-detail>